

INPATIENT DELIRIUM

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Critical Care Medicine



Objectives

- At the end of this one-hour lecture, students will be able to
 - Recognize the common clinical features of delirium
 - Identify risk factors associated with the development of delirium
 - Understand long term consequences associated with development of delirium
 - Be able to identify validated diagnostic tools used to identify patients with delirium
 - Understand principles of pharmacological and non-pharmacological treatment modalities for delirium
 - Recognize components of the ABCDEF bundle for prevention of ICU delirium

Definitions

- Terminology for delirium is confusing.
- No one universal definition exists. Different societies and organizations have their own definitions.
- There are over 30 different descriptive terms for delirium:
 - Altered mental status
 - Acute confusional state/encephalopathy
 - Toxic metabolic encephalopathy
- Overall, think of delirium as **acute brain dysfunction**

Definitions:

- Delirium: an acute change in attention, awareness, and/or cognition caused by illness that can't be explained by a pre-existing neurocognitive disorder. Often reversible.

Table 1. Diagnostic Criteria for Delirium.

Source of Criteria

DSM-5*

The presence of delirium requires all the criteria to be met:

Disturbance in attention and awareness

Disturbance develops acutely and tends to fluctuate in severity

At least one additional disturbance in cognition

Disturbances are not better explained by a preexisting dementia

Disturbances do not occur in the context of a severely reduced level of arousal or coma

Evidence of an underlying organic cause or causes

Confusion Assessment Method (CAM)†

The presence of delirium requires features 1 and 2 and either 3 or 4:

Acute change in mental status with a fluctuating course (feature 1)

Inattention (feature 2)

Disorganized thinking (feature 3)

Altered level of consciousness (feature 4)

* The criteria are adapted from the *Diagnostic and Statistical Manual of Mental Disorders*, fifth edition (DSM-5).⁵

† The criteria are adapted from Inouye et al.⁶

Types of Delirium

- Hypoactive
 - Characterized by lethargy, motor slowness, social withdrawal
 - These patients often get missed because they are not acting out
 - Associated with increased mortality, length of stay, falls, lower quality of life
- Hyperactive
 - Characterized by agitation, aggression, disorientation, hallucinations
 - These are the aggressive patients that are yelling at staff in the middle of the night!
- Mixed
 - Fluctuates between hyperactive and hypoactive

Epidemiology

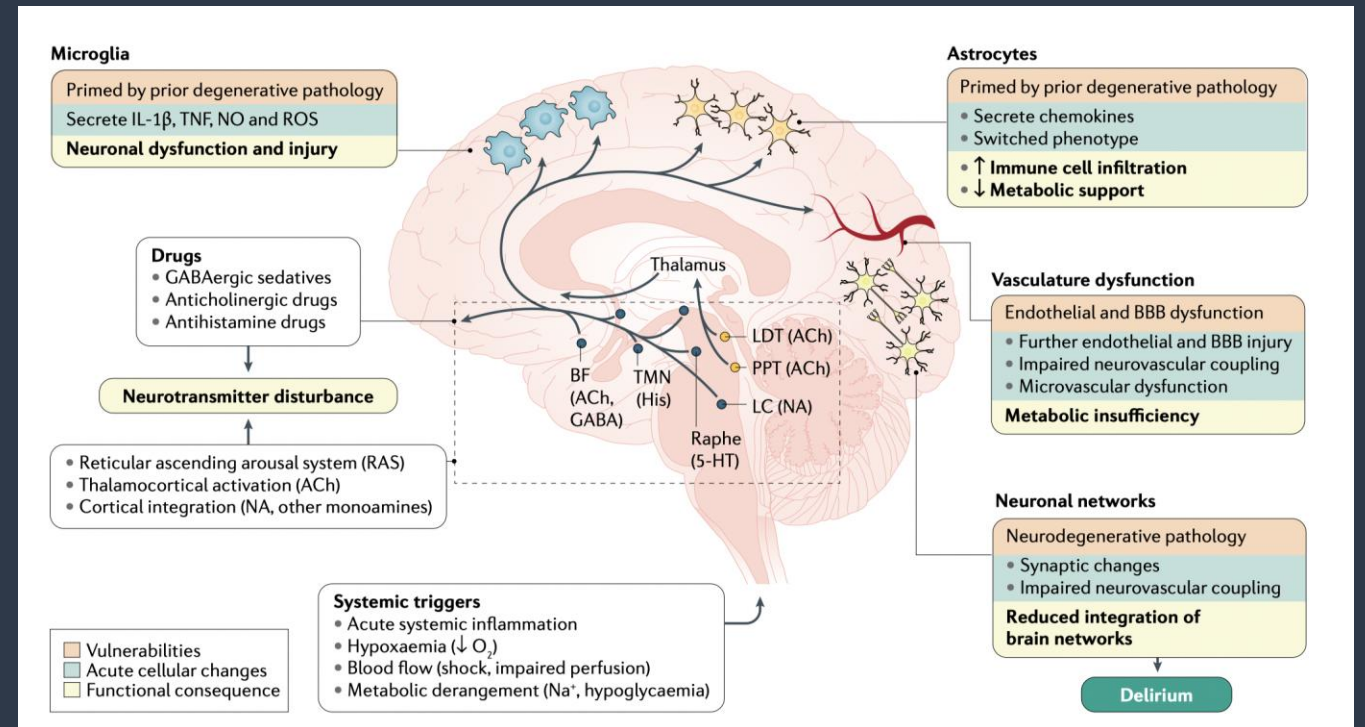
- Delirium is **COMMON**
 - Inpatient ~23%
 - Post surgical >20% in high risk patients, emergency surgery
 - Acute stroke ~25%
 - Palliative care ~60-90% at end of life
 - ICU ~30% for all ICU, ~50-70% for mechanical ventilation

Clinical Significance

- Major public health problem: persistent morbidity and **long term cognitive impairment**.
 - 20% of patients have persistent cognitive impairment at 6 months
- Associated with many **poor outcomes**:
 - Increased mortality
 - New institutionalization – discharge to destinations other than home (i.e. nursing home)
 - Dementia
 - Restrictions in motor function
- Outcomes worse in elderly, frail, hypoactive delirium, longer duration or severe delirium

Pathophysiology

- Active area of research
- Delirium usually the result of **underlying pre-disposition + acute stressor**
- Multiple etiologies contribute to development of delirium in a patient, and there are probably several neurobiological mechanisms leading to the syndrome
- Difficult to classify a single etiology or distinct mechanism
- Several theories exist...



Pathophysiology

- Brain energy metabolism:
 - Brain requires a lot of energy in the form of oxygen and glucose
 - Acute illness causing **hypoxemia or hypoglycemia** can cause bioenergetic insufficiency resulting in delirium.
 - Impaired lung function such as respiratory failure can cause hypoxemia
 - Shock can impair blood flow through microcirculatory failure resulting in relative hypoxemia and hypoglycemia
 - Sepsis results in impaired autoregulation and reduced cerebral blood flow
 - Surgery and trauma can result in impaired insulin sensitivity - impaired glucose uptake in the setting of a normal blood sugar

Pathophysiology

- Inflammation
 - Peripheral inflammation is a trigger for delirium
 - Mechanism poorly understood
 - Possible theory is that trauma, surgery, infection, sepsis trigger inflammatory mediators that can cross the blood brain barrier resulting in neuronal injury and dysfunction.

Pathophysiology

- Drugs, stress and neurotransmitter imbalance:
 - Drug use and medication changes can precipitate delirium
 - Direct effects on neurotransmitters i.e. acetylcholine, dopamine, GABA
 - **Polypharmacy** in conjunction with renal or hepatic insufficiency can lead to inappropriately high drug concentrations and adverse drug reactions
 - These 'chemical insults' can result in a theorized neurochemical basis for delirium

Clinical Presentation and Features

- **Fluctuating course:** variability is characteristic, lucid intervals are not uncommon, can be a frequent cause of confusion among clinicians.
- **Attentional deficits:** the patient is easily distracted by incidental stimuli in the environment
- **Disturbances of arousal:** the reticular activating system can be hypoactive, in which case the patient may appear apathetic, somnolent, or depressed.
- **Sleep-Wake Disturbance**
- **Impaired Memory:** especially immediate and recent memory. Following recovery patients may be amnesic for the entire episode,
- **Disorganized thinking and impaired speech**
- **Disorientation:** usually to time, often to place, but rarely to person.
- **Altered perception:** may have delusions and hallucinations. Visual hallucinations are most common.
- **Emotional disturbances:** intense emotional responses with rapid fluctuations are not uncommon.

Interactions

- Delirium is complicated and multifactorial
 - Inherently related to being sick
 - Part of a **feedback loop** that is the perfect set up for prolonged delirium
 - Treatment is to break this cycle!

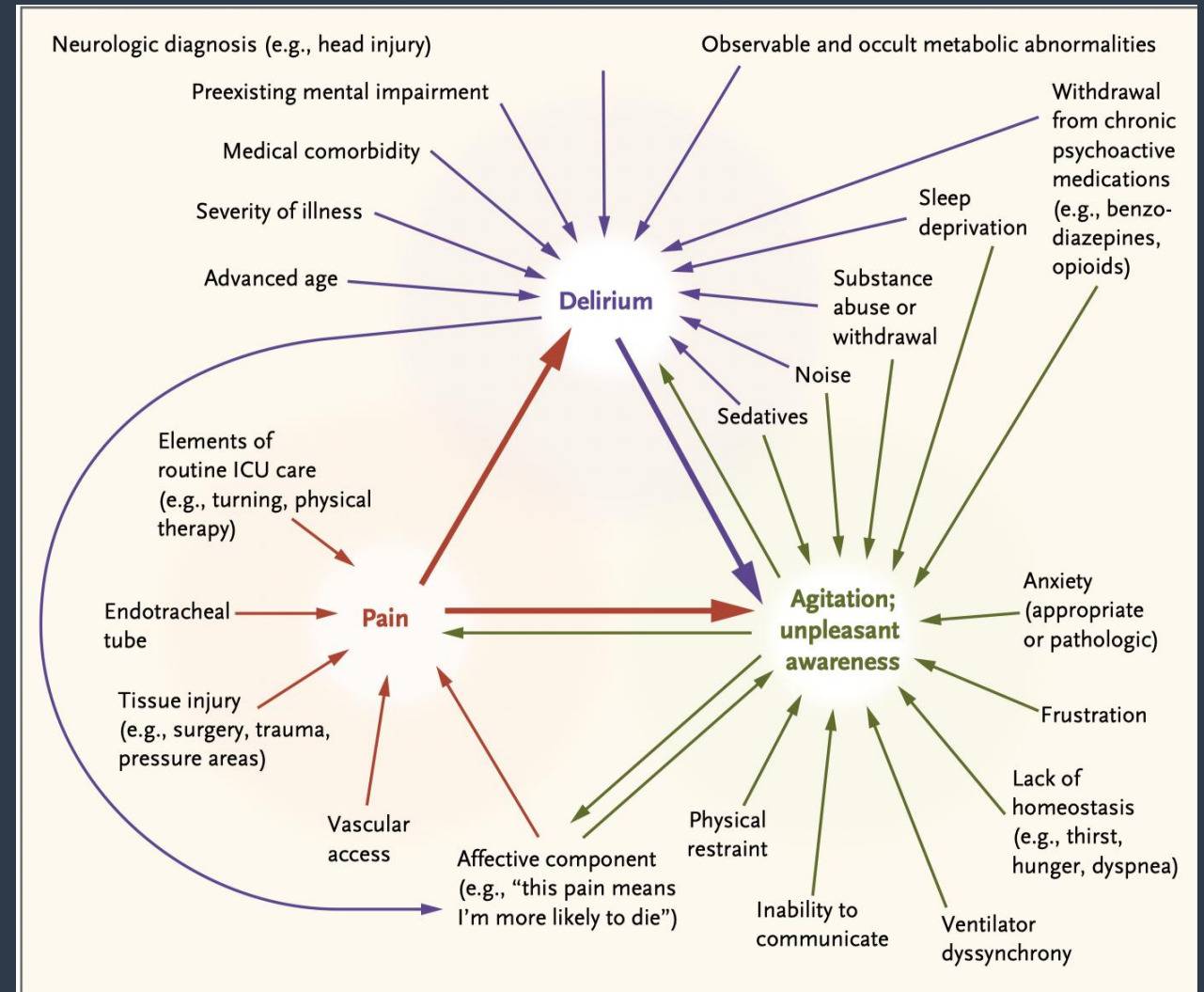
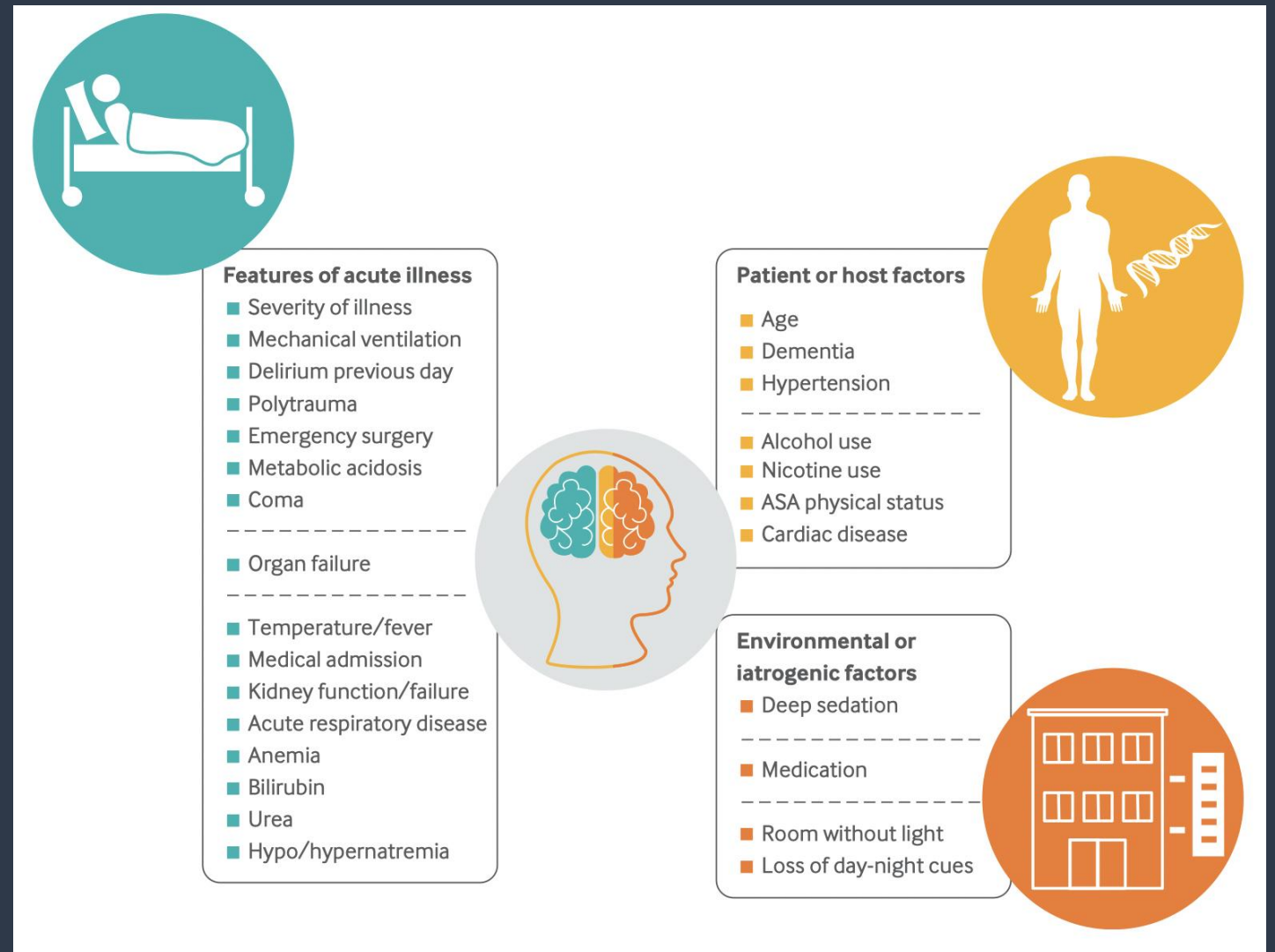


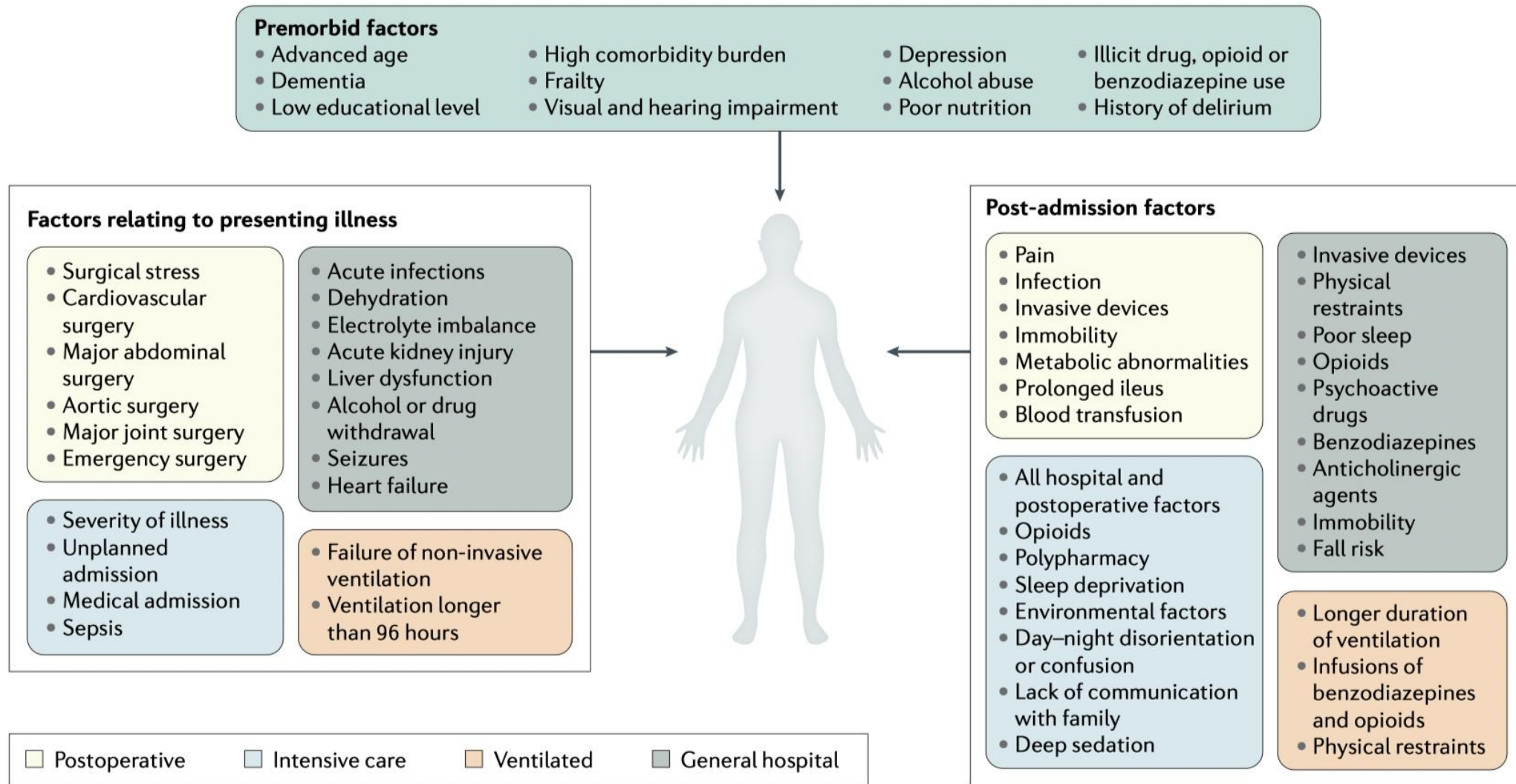
Figure 1. Causes and Interactions of Pain, Agitation, and Delirium.

Drugs and other treatments for pain, agitation, and delirium form an "ICU triad" cognitive management analogous to the "triad of anesthesia," which highlights interactions among hypnotics, analgesics, and muscle relaxants to encourage balanced anesthesia. The "ICU triad" concept highlights that changing one element is unlikely to be as effective as a coordinated approach.

Risk Factors

- **Patient** associated risk factors – generally non modifiable
- **Acute illness** associated risk factors – this is what you're trying to fix in the hospital, may or may not be modifiable
- **Iatrogenic** risk factors – these are things under your control! – more likely to be modifiable





Iatrogenic risk factors: Medications

****Beer's Criteria:**

- Widely used, developed to reduce potentially inappropriate and harmful polypharmacy in the geriatric population.
- Includes 50+ medications that should be avoided in elderly patients due to risk of adverse events
 - α -blockers (risk of hypotension)
 - Anticholinergics
 - Antidepressants
 - Antihistamines
 - Opioids
 - Benzodiazepines
 - NSAIDs
 - PPIs

Table 4. High-Risk Drugs in Delirium and Potential Substitutes.*

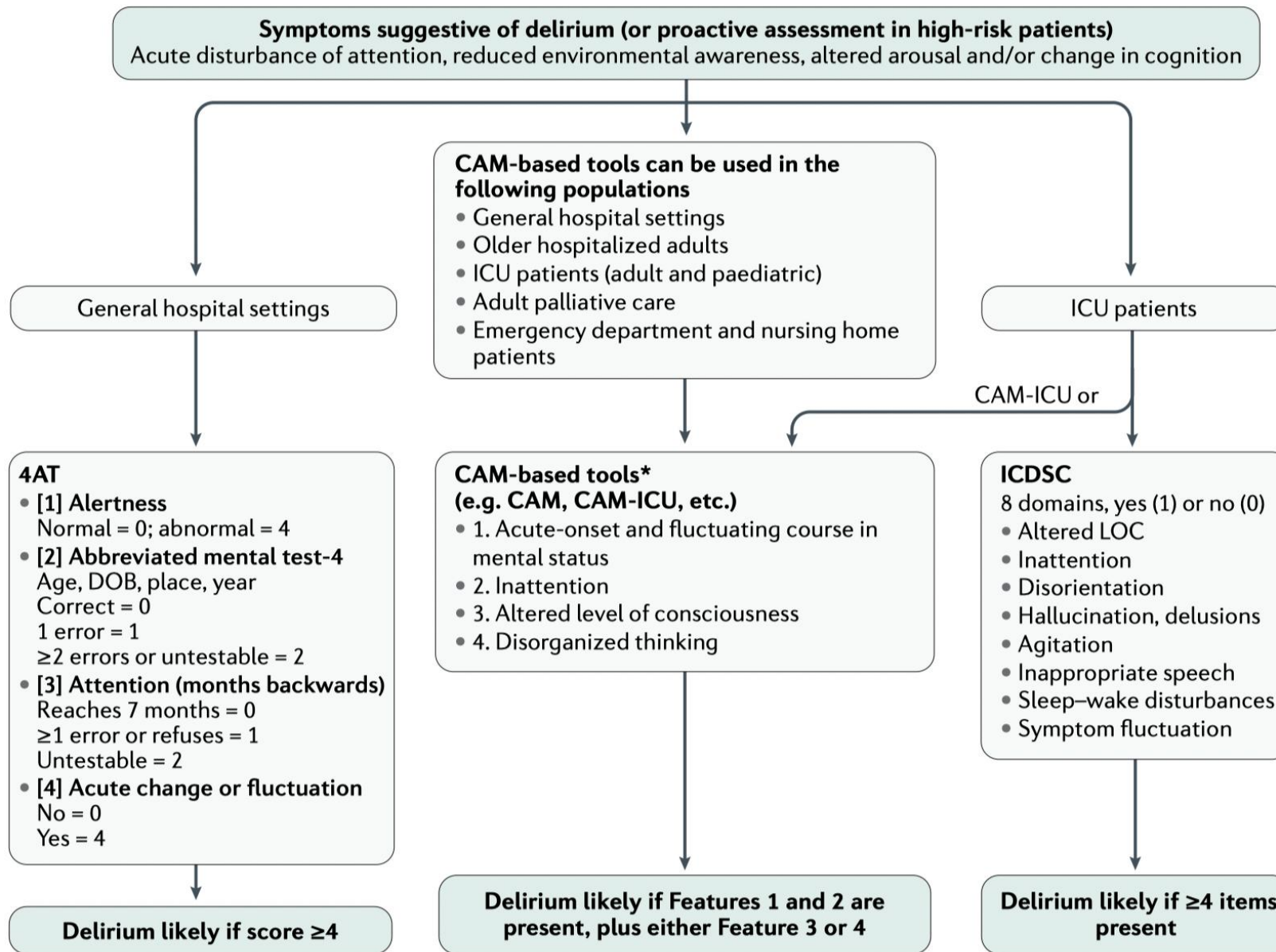
Drug	Mechanism of Adverse Effect
Benzodiazepines	CNS sedation and withdrawal
Opioid analgesics (especially meperidine)	Anticholinergic toxicity, CNS sedation, and fecal impaction
Nonbenzodiazepine sedative hypnotics (e.g., zolpidem)	CNS sedation and withdrawal
Antihistamines, especially first-generation sedating agents (e.g., doxylamine and diphenhydramine)	Anticholinergic toxicity
Alcohol	CNS sedation and withdrawal
Anticholinergics (e.g., oxybutynin and benztropine)	Anticholinergic toxicity
Anticonvulsants (e.g., primidone, phenobarbital, and phenytoin)	CNS sedation
Tricyclic antidepressants, especially tertiary amines (e.g., amitriptyline, imipramine, and doxepin)	Anticholinergic toxicity
Histamine H ₂ -receptor blockers	Anticholinergic toxicity
Antiparkinsonian agents (e.g., levodopa and amantadine)	Dopaminergic toxicity
Antipsychotics, especially low-potency typical antipsychotics (e.g., chlorpromazine and thioridazine)	Anticholinergic toxicity as well as CNS sedation
Barbiturates	CNS sedation and severe withdrawal syndrome

Evaluation of a patient with suspected delirium

- Because delirium is tied to underlying comorbidities, evaluation is broad and patient specific
- Initial assessment involves **detailed history and physical**, review of multiple possible triggers/risk factors
 - These patients will often have a non focal neurological exam – a focal deficit should alert you to other etiologies of acute encephalopathy
- May involve routine blood tests, radiological examination
 - DDx: sodium disorders, hepatic or uremic encephalopathy, intoxications, cns infections, hemorrhagic or ischemic stroke, etc
 - Potential Labs: chemistry, liver function, renal function, drug screen, cultures
- “Brain specific” tests may include: CT, MRI, EEG, lumbar puncture, antibody testing for autoimmune encephalitis etc as indicated
 - **Not all patients will need all or even any of these tests. It depends on the clinical presentation.**
 - For example: one study showed that over an 18 month period, only 11% of patients with delirium who underwent CT head had positive findings on imaging.

Assessment & Diagnosis: Validated diagnostic tools

- More than 50 delirium assessment tools have been developed, not all are validated for clinical use
- We will cover commonly used screening tools used in research and clinical settings
- General Hospital Settings
 - 4 As Test or Confusion Assessment Method (CAM)
 - CAM is the most commonly used tool in research and clinical practice
- ICU patients
 - CAM ICU or Intensive Care Delirium Screening Checklist (ICDSC)
 - CAM ICU sensitivity 80%, specificity 96%
 - ICDSC sensitivity 74%, specificity 82%
 - CAM ICU is specifically designed for critically ill patients and those on mechanical ventilation



Management: An ounce of prevention...

- Prevention is key!
 - Non pharmacological interventions
 - Be on the look out for risk factors
 - Stop offending medications: **beware of benzodiazepines**
 - Don't torture the patient
 - Pharmacological interventions
 - Multiple studies have looked at use of antipsychotic drugs in high risk patients
 - **Evidence does not support use of haloperidol or second generation antipsychotics in prevention of delirium**

Box 2 | Delirium prevention in different health-care settings

Consensus guidelines^{241,262} make a number of recommendations for delirium prevention in various health-care settings.

General settings

Multicomponent interventions

- Early recognition of high-risk factors (age >65 years, dementia, hip surgery and high acuity)
- Daily screening for delirium
- Environmental orientation (sensory, auditory, dentures, time, events, family visits and music)
- Maintain normal hydration
- Regulation of bladder and bowel function
- Early establishment of normal diet
- Correction of metabolic disorders
- Cardiorespiratory optimization (with provision of oxygen if appropriate)
- Early identification of infection
- Effective treatment of pain
- Daily mobilization
- Avoidance of antipsychotic drugs
- Avoidance of benzodiazepines
- Reduced nocturnal disturbances to promote sleep

- Early removal of devices (intravascular and airway devices)
- Avoidance of physical restraints
- Sleep promotion (eye mask and earplugs)

Pharmacological interventions

- None with high-level evidence

Intensive care and high acuity units, intubated and non-intubated patients

Non-pharmacological interventions (as above, consider also)

- Early recognition of high-risk patients (age >65 years, high acuity, sepsis, shock, dementia and ventilation)
- Light sedation
- No benzodiazepines
- Early mobilization
- Promotion of day–night routine
- Environmental awareness and orientation
- Removal of devices (intravascular and airway devices)

Pharmacological interventions

- Some suggestive evidence but not recommended by consensus guidelines

Management: Treatment

- **Multidomain approach:** not one size fits all, emphasis on **non-pharmacological** treatments
 - Addressing multiple delirium triggers
 - Correcting physiological disturbances
 - Treating symptoms of delirium such as distress
 - Communicating with patients and caregivers
 - Addressing current and future risks linked with delirium
- **Pharmacological approach** is not supported by current guidelines
 - No evidence that or consensus that delirium should be treated with one or more drugs
 - This doesn't stop us from using first or second generation antipsychotics in the inpatient setting. Most commonly used drugs are haloperidol, quetiapine, risperidone, and olanzapine in my experience...they work poorly

Treatment of delirium in the ICU: The ABCDEF Bundle

- ICU is a deliriogenic environment – most patients are exposed to 10+ delirium risk factors during their critical illness
 - Advanced age, vision/hearing impairment, immobility, pain, hypovolemia, electrolyte imbalances etc.
- Research shows that implementation of bundled strategies can reduce delirium in the ICU
- ABCDEF bundle:** promotes awake, spontaneously breathing, mobile patients
 - A**ssess/prevent/manage PAIN
 - B**oth SAT/SBT
 - C**hoice of sedation and analgesia
 - D**elirium – use a screening tool daily
 - E**arly mobility
 - F**amily engagement

Table 1 | Delirium risk factors targeted by the ABCDEF bundle

ABCDEF bundle component ²⁶⁷	Approach	Delirium risk factor
A	Assess, prevent and manage pain	Undertreated pain; over-sedation caused by analgesics
B	Both spontaneous awakening trials and spontaneous breathing trials	Over-sedation; mechanical ventilation
C	Choice of sedation and analgesia	Over-sedation caused by analgesics; over-sedation caused by sedatives; exposure to deliriogenic sedatives (for example, benzodiazepines)
D	Delirium: assess, prevent and manage	Disordered sleep–wake cycle; vision and hearing impairment; other unrecognized delirium risk factors
E	Early mobility and exercise	Immobility
F	Family engagement and empowerment	Disorientation

Delirium and Dementia

- Delirium and dementia are **DIFFERENT** disease processes. **They are not on a spectrum or continuum. Not a causal relationship in either direction.**
- A patient can have delirium **WITHOUT** pre-existing dementia. Patients with delirium do not have dementia automatically.
- Patients who develop delirium may have higher risk of cognitive impairment in the future, but it is not necessary that they will go on to develop dementia.
- Patients with pre-existing dementia may have a higher chance of developing delirium in the inpatient setting
- Delirium is an **acute reversible** disease process
- Dementia is a **chronic and progressive** neurocognitive illness
- Please refer to Dr. Massa and Dr. Klapstein's lectures on dementia for further information

The Bottom Line

- Delirium is common
- Delirium is associated with poor short and long term outcomes, increased morbidity and mortality, and overall poor quality of life
- Use validated screening tools when you suspect delirium
- Prevention of delirium is of utmost importance
- Pharmacological interventions don't work in prevention or treatment of delirium
- Don't torture the patient

References

- Marcantonio, Edward R. 2017. “Delirium in Hospitalized Older Adults.” *The New England Journal of Medicine* 377 (15): 1456–66.
- Reade, Michael C., and Simon Finfer. 2014. “Sedation and Delirium in the Intensive Care Unit.” *The New England Journal of Medicine* 370 (5): 444–54.
- Wilcox, M. Elizabeth, Timothy D. Girard, and Catherine L. Hough. 2021. “Delirium and Long Term Cognition in Critically Ill Patients.” *BMJ* 373 (June): n1007.
- Wilson, Jo Ellen, Matthew F. Mart, Colm Cunningham, Yahya Shehabi, Timothy D. Girard, Alasdair M. J. MacLulich, Arjen J. C. Slooter, and E. Wesley Ely. 2020. “Delirium.” *Nature Reviews. Disease Primers* 6 (1): 90.

Questions

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